Steps to follow for Digital Assignment -2

1. Should have MATLAB 2025b or later versions as it requires RL ToolBox and Deep Learning ToolBox.
2. Open the following official MathWorks page: (all info given in this page) https://in.mathworks.com/help/releases/R2025b/reinforcement-learning/ug/tune-pi-controller-using-td3.html?utm_source=chatgpt.com

3. Open MATLAB

   Open in MATLAB Online or **Open Live Script**
   depending on the MATLAB version/interface.
   The example is titled: **Tune PI Controller Using Reinforcement Learning**
   It uses a water-tank control system and the **TD3 reinforcement-learning algorithm**.

4. Command for uploading models in Comamnd windows (Given in Link also)

```
Model - rlwatertankPIDTune.slx
pretrained TD3 agent - WaterTankPIDtd3.mat
supporting MATLAB code - myBasisFcn.m
pretrained agent - load("WaterTankPIDtd3.mat","agent")
```

5. Use same plant model, replace PI controller with LSTM considering previous 4 memory samples.

6. **Result: comparison of [Desired water level – feed back] in highlighted area error output signal with conventional PI conroller, TD3-RL and LSTM.**

